Suggested problems: 1-13 odd, 17-29 odd

Problem 1: The price P in dollars to purchase a used car is a function of its original cost, C, in dollars, and its age, A, in years.

(a) What are the units of $\partial P/\partial A$?

The units are dollar per year.

(b) What is the sign of $\partial P/\partial A$ and why?

The sign should be negative. This is because as a car ages, its value goes down.

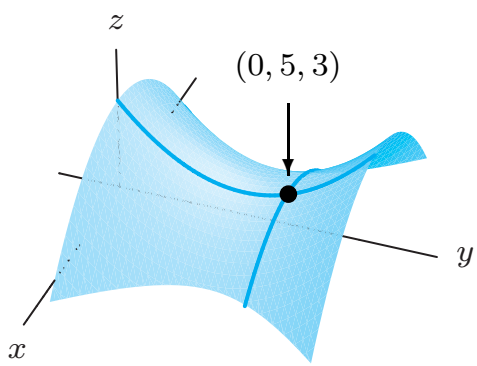
(c) What are the units of $\partial P/\partial C$?

The units are dollar per dollar.

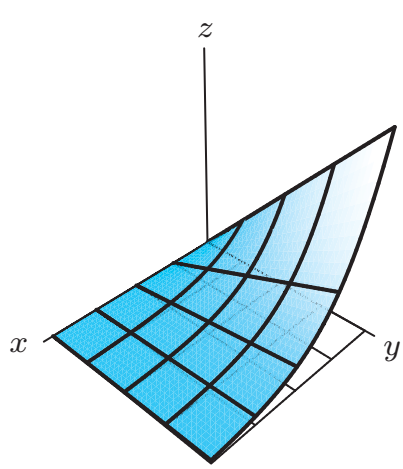
(d) What is the sign of $\partial P/\partial C$ and why?

The sign should be positive. This is because a car with a higher original cost should be more expensive to buy used than a car with a cheaper original cost of the same age.

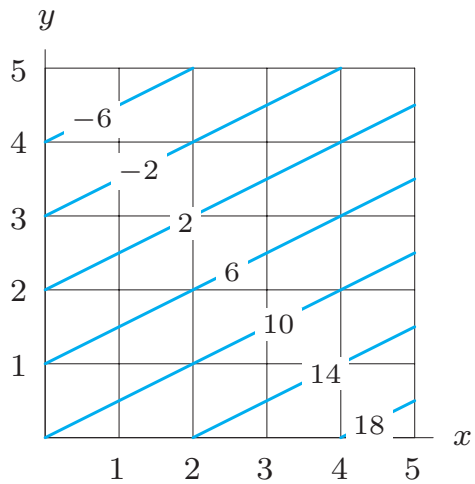
Problem 2: The figure below shows the surface $z = f(x, y)$. What is the sign of $f_x(0, 5)$? What is the sign of $f_y(0, 5)$? We have $f_x(0, 5) < 0$ and $f_y(0, 5) > 0$.



Problem 3: The figure below shows the graph of a function $x = f(x, y)$ on the domain $0 \le x \le 4$ and $0 \le y \le 4$. Use the graph to rank the following quantities in order from smallest to largest: $f_x(3, 2)$, $f_x(1, 2)$, $f_y(3, 2)$, $f_y(1, 2)$, 0. $f_x(1, 2) < f_x(3, 2) < 0 < f_y(3, 2) < f_y(1, 2)$



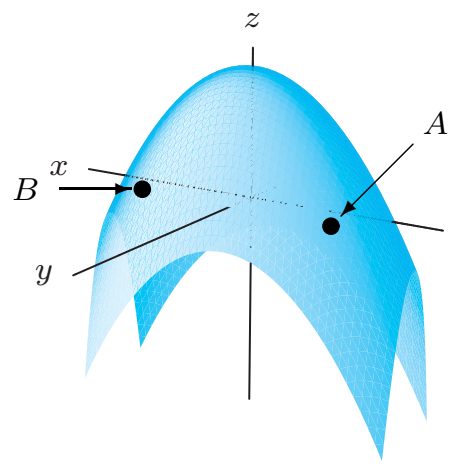
Problem 4: Suppose we are given the contour diagram below of a function $z = f(x, y)$. Is f_x positive or negative? How about f_y? $f_x > 0$ and $f_y < 0$.



Problem 5: The graph of $z = f(x, y)$ is shown in the figure below. The points A and B are in the xy-plane.

(a) What is the sign of $f_x(A)$? What is the sign of $f_y(A)$?

(b) The point P in the xy-plane moves along a straight line from A to B. How does the sign of $f_x(P)$ change? How does the sign of $f_y(P)$ change?



(a) $f_x(A) > 0$ and $f_y(A) < 0$. (b) The sign of $f_x(P)$ appears to be positive until the point P crosses the y-axis, and then it turns negative. The sign of $f_y(P)$ appears to be negative at all times as P moves from A to B.